

inference engineering weekly assignment

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0.1 Question 1 What is the difference between latency and throughput in model inference? Which one influences performance more, and how?

latency is the time taken for the LLM to respond, throughput is the number of requests the LLM model can handle at the same time. the factor that will more affect the model's performance is throughput, it should be able to handle all the requests efficiently, if it can't it would cause latency due to piling up of requests and it might not answer completely. Hence throughput affects the model's output the most

0.2 2. Explain KV cache. How does it help model generation?

KV cache is the short memory used by LLM to store the information generated from the previous pairs. This is necessary because it saves the computational overhead from calculating the values again. as the model stores the information, it doesn't need to calculate those values again.

0.3 3. What is quantization? Explain the accuracy trade-off between INT8 and INT4. Include an example with calculations/formula.

Quantization is the process of representing model values using fewer bits. For example, a model can use INT8 or INT4 instead of higher-precision formats such as FP32.

INT8 uses 8 bits to represent a value, while INT4 uses only 4 bits. This reduces the memory required by the model and can also improve inference speed. However, using fewer bits can reduce the accuracy of the model.

For example, suppose the original value is 10, the scale is 0.1, and the zero-point is 0.

The quantized value can be calculated as:

$$q = \frac{x}{scale} + zero_point$$

Substituting the values:

$$q = \frac{10}{0.1} + 0$$

$$q = 100$$

Therefore, the original value 10 is represented by the quantized value 100.

INT4 generally requires less memory than INT8, but it can have a higher chance of accuracy loss. Therefore, quantization involves a trade-off between model size, inference speed, and accuracy.

0.4 4. Name one inference serving framework (e.g., vLLM or TensorRT-LLM) and briefly describe one optimization.

One example of an inference serving framework is vLLM. It is used to efficiently serve large language models and handle multiple inference requests.

One optimization used by vLLM is PagedAttention. It manages the KV cache in smaller blocks of memory instead of requiring one large continuous memory area. This helps use memory more efficiently when multiple requests are being processed.